

Vanilla vs LUT47: what changes inside the model when the MLP FFN becomes a LUT

ffn_replacement interpretability analysis

September 24, 2026

Models (both MinimalGPT, $E=384$, attention $H=6$, depth 6, RoPE, ClimMix, same tokenizer):

- **A) Vanilla** — standard dense-MLP FFN. Run `exp_n_0216_vanilla48k_gelu_ckpt2k_seed1`, val_bpb **1.1151**. (The task named `exp_n_0177_vanilla48k_corrected`, but that folder has *no* checkpoint; 0216 is the substitute — a 48K vanilla with an essentially identical val_bpb 1.115094 vs 1.115092 — and it has weights saved.)
- **B) LUT47** — FFN replaced by a light CompressionMultiHeadLUT (H8/tph64, nap8, read_top_n=2, float read, dropout+TV). Run `exp_n_abl_47_B48k_light_lmfrozeng_n2_tau0p5_tv10_tph64_h8_seed1_n`, val_bpb **1.1079** (beats vanilla by 0.0072).

Both checkpoints load with missing=0/unexpected=0 and run forward passes; all activations captured via forward hooks on the identical attention path and on the FFN slot. Shared eval batch: 8×256 real ClimMix val tokens, same tokens to both models.

1 Attention patterns

The attention *architecture* is identical, so any systematic difference is **demand**ed by the FFN. The LUT model’s attention is systematically **sharper** (lower entropy): mean 3.10 bits (LUT) vs 3.60 (vanilla), sharpest at layers 1/2/5. Layer 0 is diffuse in both (~ 5.7 bits). Induction heads exist in both at comparable strength (~ 0.14), slightly relocated (vanilla L3H1, LUT L4H0); max previous-token attention 0.575 vs 0.487. *Takeaway*: a discrete/low-rank LUT read wants cleaner, more decisive inputs, so the shared attention learns to be sharper.

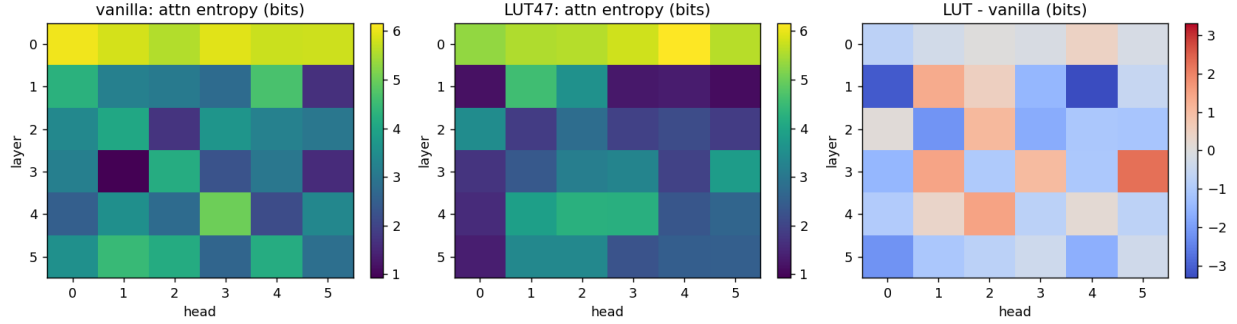


Figure 1: Per-(layer, head) mean attention entropy (bits): vanilla, LUT, and their difference. Blue in the diff = the LUT sharpens that head.

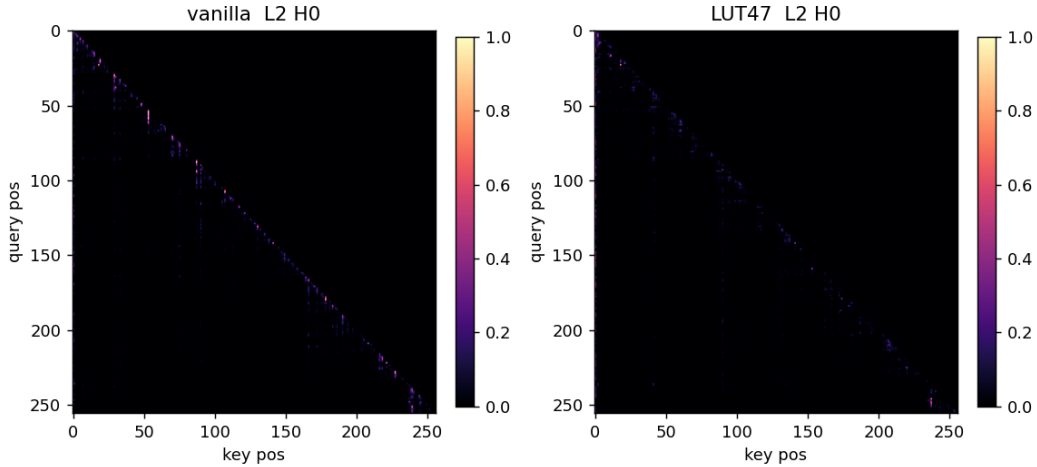


Figure 2: Example attention heatmap (layer 2, head 0) on the shared batch.

2 Embeddings

Separately trained, so absolute directions do not align — comparing *structure*. Token embeddings are near-full-rank and near-isotropic in both (effective rank ≈ 342 (vanilla) vs 333 (LUT) out of 384; isotropy 0.14 vs 0.17). The untied unembedder is **low-rank and highly anisotropic in both, and nearly identical** (effective rank ≈ 63 , isotropy ≈ 0.89 — the usual dominant “rogue direction”). *Takeaway*: swapping the FFN barely touches the embedding/unembedding geometry.

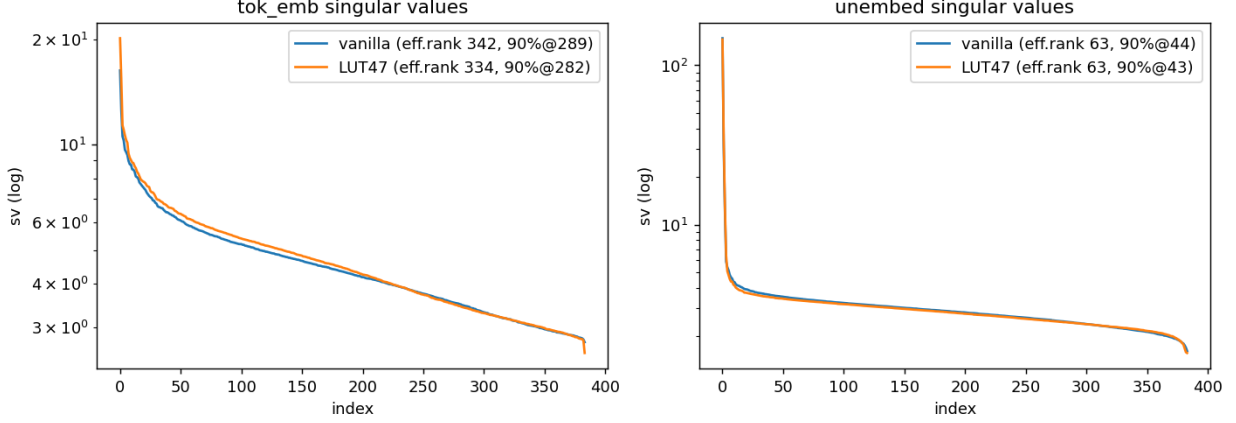


Figure 3: Singular-value spectra (log) of the token embedding and the untied unembedder, with effective rank. The unembedder is far lower-rank than the token embedding and is essentially the same for both models.

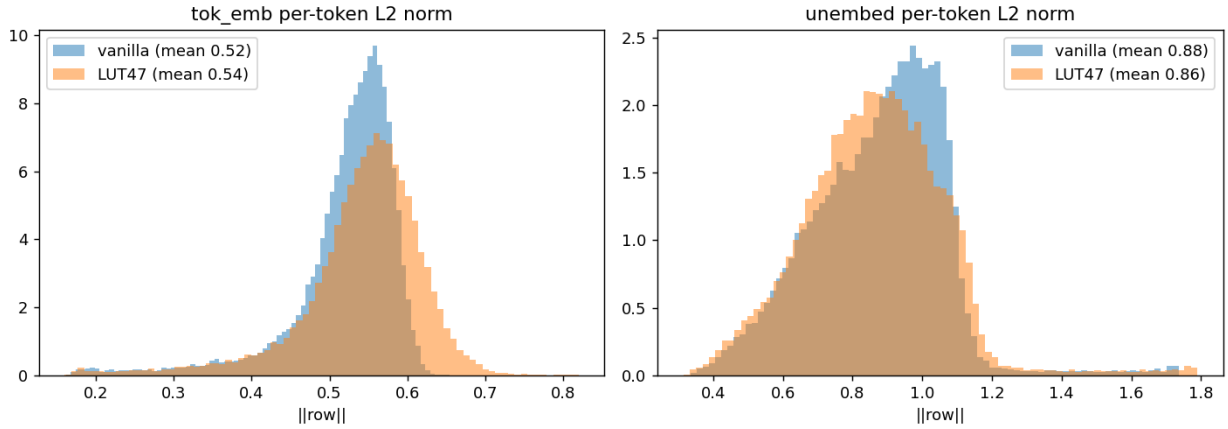


Figure 4: Per-token L2-norm distributions for the token embedding and unembedder.

3 FFN vs LUT internals — the biggest differences

The LUT FFN writes **much smaller residual updates** and the model runs at a **smaller residual scale**: mean $\|\text{FFN } \Delta\|$ per layer vanilla [5.5, 3.3, 3.6, 4.5, 5.9, 12.8] vs LUT [0.6, 1.6, 1.1, 1.3, 1.9, 5.4]; residual norm up to 18.2 (vanilla) vs 9.9 (LUT); ratio $\Delta/\text{residual} \approx 0.4\text{--}0.95$ (vanilla) vs $0.2\text{--}0.55$ (LUT). The LUT updates are also **low-rank, especially early**: effective rank [5.5, 7.4, 188, 194, 184, 24] (LUT) vs [225, 111, 261, 259, 249, 139] (vanilla) — the early LUT layers write into a $\sim 5\text{--}7$ dimensional subspace, the signature of a discrete lookup. The LUT read is sparser than the dense GELU hidden, most at layer 0 (16.6% near-zero vs $<1\%$). *Takeaway*: the LUT is a lighter, lower-rank, sparser operator than the MLP; it moves the residual less and in fewer directions.

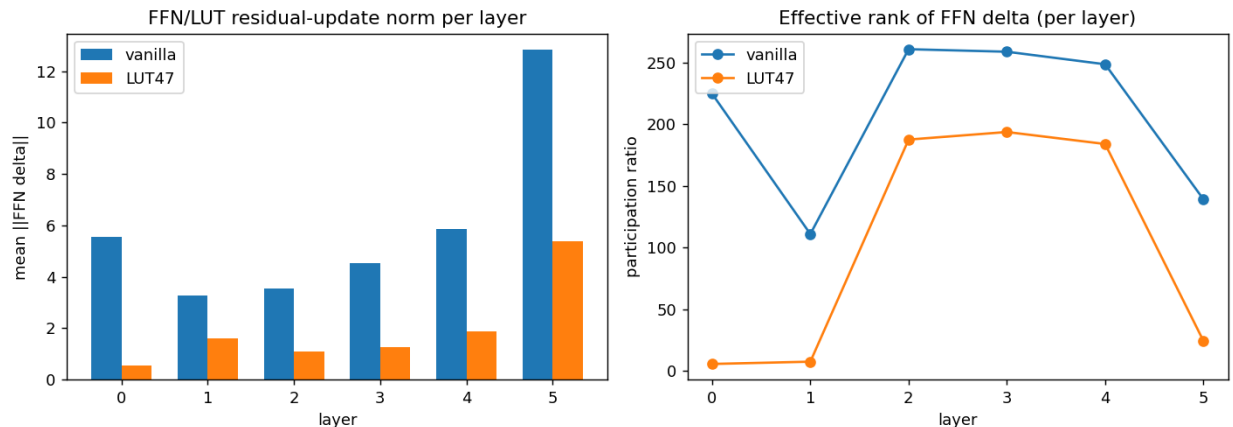


Figure 5: Left: per-layer FFN residual-update norm (the LUT writes far less than the MLP). Right: effective rank of the FFN update — the LUT’s early layers are strikingly low-rank.

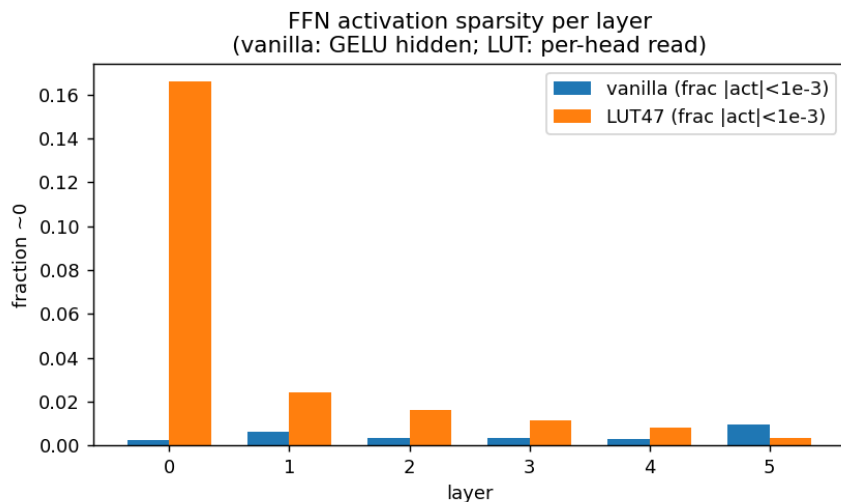


Figure 6: Per-layer FFN activation sparsity (fraction near zero): the LUT read is sparser early.

4 Single-sentence trace

Side-by-side on `The capital of France is` and `The cat sat on the mat.`: next-token top-5, per-layer attn/FFN/residual norms, and per-head attention (full detail in `sentence_trace.md`). It confirms the aggregate picture: on the France prompt the LUT FFN’s final-layer update is 2.38 vs the vanilla 10.38 (Δ /residual 0.19 vs 0.53), and the LUT’s layer-5 heads attend ~ 0.90 to the last token while the vanilla heads spread across ‘is’/‘of’/‘capital’. Neither small ClimbMix model actually predicts “Paris” — both are weak; this is about mechanism, not factual recall.

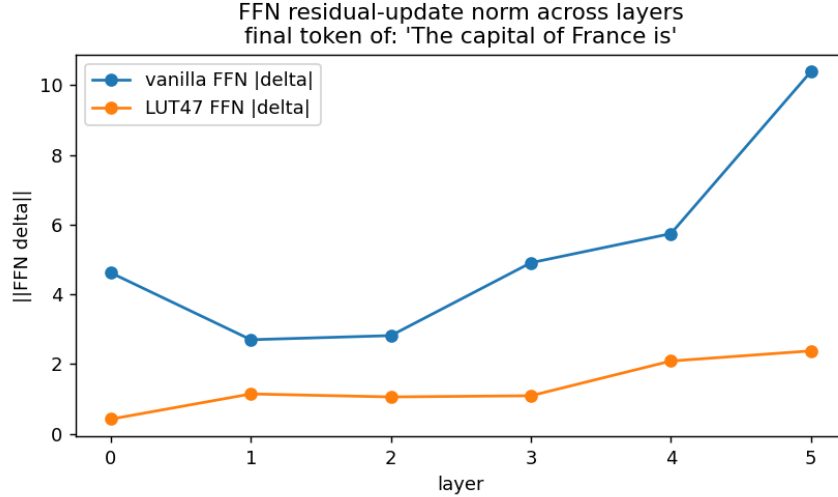


Figure 7: FFN residual-update norm across layers on the final token of “The capital of France is”.

5 Per-head LightMHL FFN characterization (LUT47, H8/tph64)

Each LightMHL head of the FFN LUT is a map $\mathbb{R}^{48} \rightarrow \mathbb{R}^{48}$ (its 48-dim input code slice $\rightarrow$ its 48-dim top- n -blended read, *before* decompress). Confirmed dims: 6 layers $\times$ 8 heads, tph=64 tables/head, nap=8 ($2^8=256$ cells/table), input_dim/head = output_dim/head = 48. Driven on 16×256 real ClimbMix tokens; cell selections captured by disabling the compiled path and wrapping `_pack_index`.

Findings.

- **Codebook usage is DENSE, not sparse.** Across the batch $\sim 16,000$ of the 16,384 possible (table, cell) selections fire ($\sim 97\%$), with cell-selection entropy ~ 13.4 of a max 14 bits. It is *not* “a few cells fire”: nearly the whole codebook is exercised; the moderate output rank comes from cell *values* clustering, not from sparse addressing.
- **Per-head output rank is $\cap$ -shaped with depth:** mean effective rank 27 (L0) $\rightarrow$ 46–47 (L2–4, near-full 48) $\rightarrow$ 38 (L5). Note this is *per head before decompress*; the full residual write collapses much lower (~ 5 –7 early, §3) because the learned dense decompress projects the correlated head outputs into a low-dim subspace at early layers — a learned collapse, not a structural cap (H8 has $8 \times 48 = 384$, no bottleneck).
- **Not single-table piecewise-constant.** A single table’s selected cell explains only 11–29% of the head-output variance (each head sums 64 tables plus a top-2 blend), so the read is a dense sum of many discrete selections rather than one switch.
- **Magnitude grows with depth,** sparsity falls: per-head $\|\text{output}\|$ 0.08 $\rightarrow$ 2.3 from L0 to L5; near-zero read fraction 16.6% (L0) $\rightarrow$ 0.3% (L5).
- **Heterogeneity at the ends.** Early (L0) and late (L5) layers contain some *near-dead* heads (output rank ~ 2 –4, tiny norm) alongside rich ones (rank ~ 38 –45); the middle layers (L2–4)

are uniformly rich (rank ~ 46 for all 8 heads). Specialization/dead heads concentrate at the ends; the middle is a uniform, high-rank workhorse.

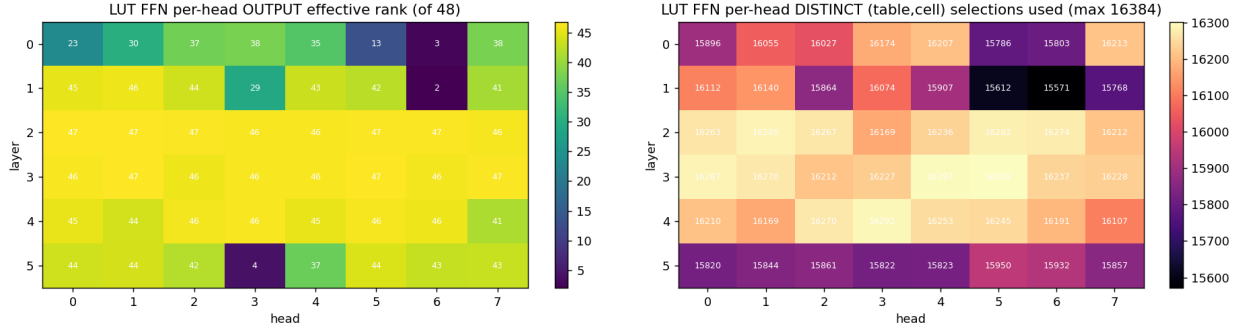


Figure 8: Per-(layer,head): output effective rank (left; note the near-dead heads at L0/L5 and the uniform middle) and distinct (table,cell) selections used (right; near-saturated $\sim 16k/16,384$ everywhere = dense codebook usage).

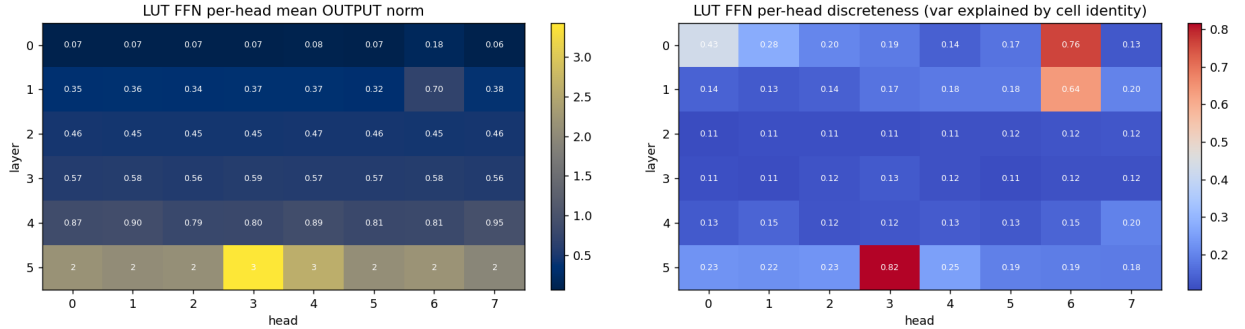


Figure 9: Per-(layer,head): mean output norm (left; grows with depth) and discreteness — fraction of output variance explained by one table's cell identity (right; low, i.e. the read is a dense sum of many selections).

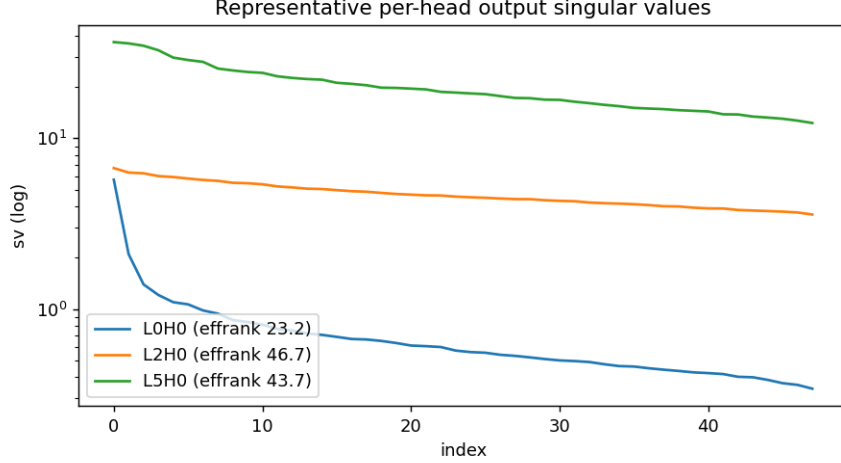


Figure 10: Representative per-head output singular-value spectra (L0/L2/L5, head 0).

5.1 Why the early residual write collapses to rank ~ 5 –7

The FFN residual write is $\text{write} = W_{\text{dec}} \cdot (\text{concat of the 8 head reads})$. For this H8 model the concatenated code is $8 \times 48 = 384$ -dim and W_{dec} is a full dense 384×384 (no structural bottleneck). We decomposed the write rank per layer into: rank of the joint 384-d code, (stable) rank of W_{dec} , rank of the write, and cross-head read correlation.

L	code eff.rank	code r99	write eff.rank	write r99	W_{dec} eff.rank	W_{dec} stable	W_{dec} r99	mean off-diag $ \cos $
0	19.4	98	5.6	5	218	20.9	259	0.115
1	21.6	148	7.3	13	245	35.6	276	0.117
2	344.7	371	193.8	246	242	54.7	274	0.116
3	344.4	372	199.9	248	243	68.6	274	0.116
4	323.1	370	188.0	243	251	70.3	278	0.116
5	61.0	274	25.6	102	259	45.7	285	0.116

Mechanism. The collapse is driven mainly by the **joint code**, not by W_{dec} : at L0–L1 the concatenated 384-d head-read code has effective rank only ~ 19 –22 (99% energy in ~ 98 –148 dims), even though each head’s 48-d read is individually rank ~ 27 –46. The 8 heads are *not* strongly pairwise-correlated (mean off-diagonal read $|\cos| \approx 0.12$ at every layer) — the redundancy is a *shared low-dimensional structure* across heads, not pairwise duplication. W_{dec} is near-full-rank at every layer (r99 ~ 274 –285), so it is **not** a structural bottleneck; but it is anisotropic, and early its *stable* rank is low (~ 21), which *compounds* the collapse — it concentrates the already-low-rank code down from ~ 20 to the ~ 5 –6-dim write. Middle layers (L2–4) do not collapse because the joint code becomes near-full-rank (~ 323 –345) and W_{dec} ’s stable rank rises (~ 55 –70). **Why early vs middle:** early layers haven’t yet built a rich residual, so the LUT reads have little to differentiate on — the joint code lives in a ~ 20 -d subspace and the FFN barely moves the residual (§3); by the middle the residual carries rich information, the reads span the full space, and the write is rich. L5 partially re-collapses (code 61, write 26) as the network narrows toward the output.

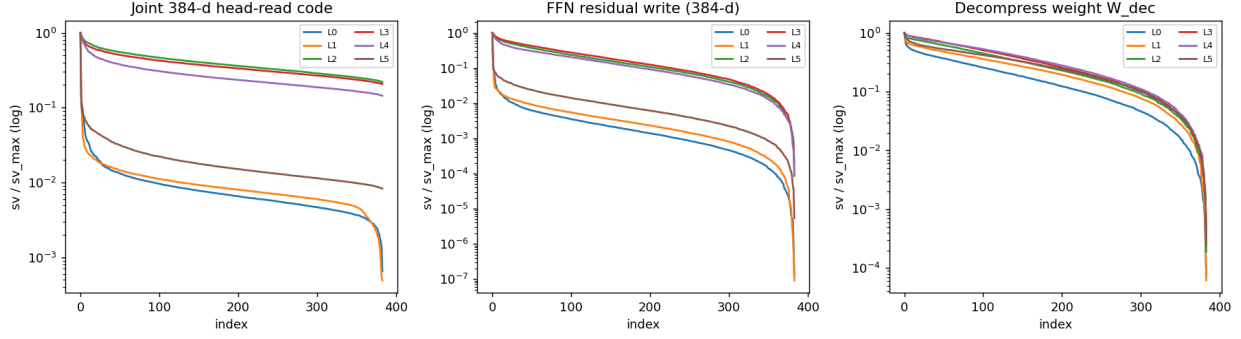


Figure 11: Per-layer normalized singular-value spectra: joint 384-d head-read code (left), the 384-d FFN write (middle), and the decompress weight W_{dec} (right). The code and write spectra drop off a cliff at L0/L1 (low rank) and are full at L2–4; W_{dec} is near-full-rank at all layers.

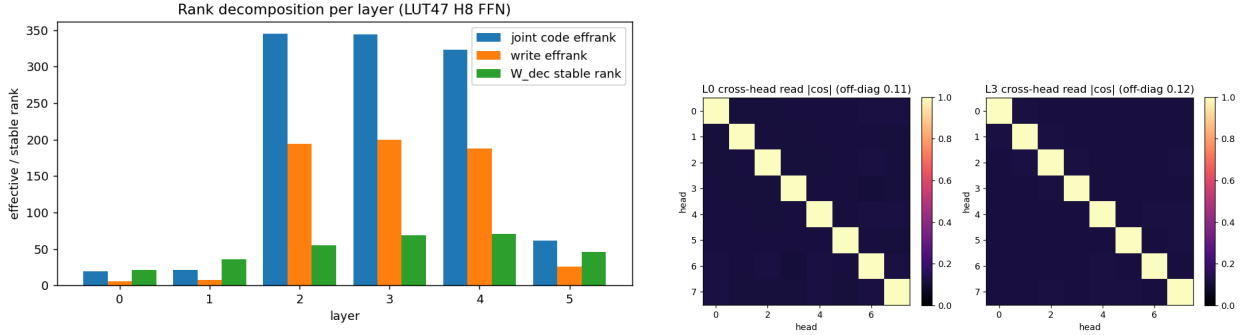


Figure 12: Left: joint-code effective rank, write effective rank, and W_{dec} stable rank per layer — the write tracks the JOINT CODE, not W_{dec} . Right: cross-head read $|\cos|$ heatmaps at the collapsed L0 vs the rich L3 — both low and similar (~ 0.12 off-diagonal), i.e. the collapse is not pairwise head correlation.

5.2 Per-table structure: is the early low rank baked into the tables?

Each head has 64 tables; each table stores $2^8=256$ cells of $d_{\text{out}}=48$ value vectors (constant cells). We compare the STATIC structure (what the stored values can span) with the DATA-DRIVEN structure (what actually fires), per layer:

L	per-table rank ($\pm$ sd, of 48)	within-head cross-table overlap	STATIC union rank	DATA output rank
0	42.0 ± 0.5	0.34	45.6	27.1
1	42.1 ± 0.5	0.30	46.4	36.5
2	44.6 ± 0.3	0.26	47.7	46.6
3	44.6 ± 0.3	0.25	47.7	46.4
4	44.3 ± 0.3	0.25	47.7	45.1
5	43.9 ± 0.3	0.24	47.7	37.6

Answer: the early low rank is **DATA-DRIVEN**, not baked into the tables.

- Each table is INDIVIDUALLY HIGH-RANK — effective rank ~ 42 – 45 of 48 at every layer. Not low-rank tables.
- Tables within a head share only MODERATE subspace overlap (0.34 early $\rightarrow 0.24$ late, where 0 = orthogonal, 1 = identical). They do *not* collapse onto the same few directions.
- STATICALLY, the 64 tables jointly span ~ 46 – 48 of the 48 dims at EVERY layer (union rank), including L0 — the codebooks are rich everywhere.
- Yet the DATA-DRIVEN per-head output is only 27 at L0 vs a static span of 46 . The gap is the whole story: the early-layer inputs select cells whose *realized* (top-2-blended, small-magnitude) reads span a lower-dim subspace, even though the tables *could* produce ~ 46 . By L2–4 the data exercises the full static span ($46.6 \approx 47.7$). So the $\cap$ -shaped output rank is a property of WHICH cells fire on real tokens, consistent with the collapse analysis (early residual is thin, so the reads have little to differentiate) — it is *not* that each table is low-rank or that all tables share a few directions.

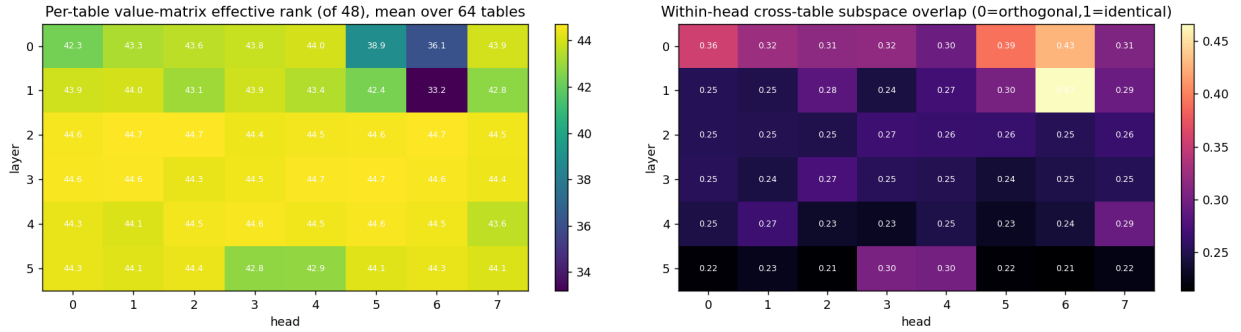


Figure 13: Per-(layer,head): mean per-table value-matrix rank (left; high, ~ 42 – 45 everywhere) and within-head cross-table subspace overlap (right; moderate, ~ 0.24 – 0.34 — tables are not redundant).

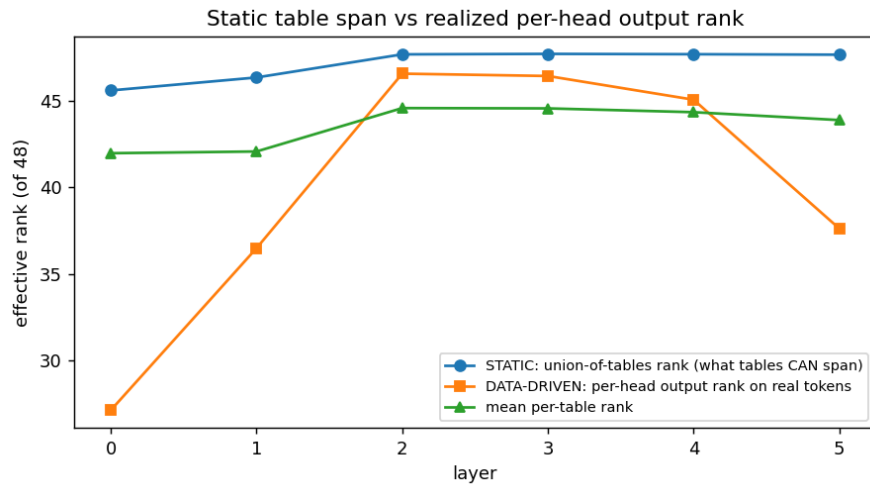


Figure 14: Static union-of-tables span (~ 46 – 48 at all layers) vs the realized per-head output rank on real tokens ($\cap$ -shaped $27 \rightarrow 46 \rightarrow 38$). The large L0/L1/L5 gap = the low rank is data-driven, not a table property.

6 Overall

Replacing the MLP with a LUT FFN **shifts work out of the FFN** (smaller, lower-rank, sparser residual writes) and **demands sharper attention**, while leaving the token/unembedding geometry essentially unchanged — and it slightly **wins** on val_bpb (1.1079 vs 1.1151). The LUT behaves like a decisive, low-rank router that offloads mixing onto a crisper attention path.